Local
Intrinsic Global
Environmental Global

Random Forest

XGBoost

All Centrals

High-Mass Satellites

Low-Mass Satellites

Relative Importance

1.0
0.8
0.6
0.4
0.2
0.0

M_{BH} Σ^* r/R_e M^* δ_5 M_h

Train AUC = 0.83
Test AUC = 0.82
Local
Env. (4%)
17%
79%
Intr.

Train AUC = 0.84
Test AUC = 0.82
Local
Env.
21%
16%
63%
Intr.

Train AUC = 0.77
Test AUC = 0.74
Env. (0%)
Local
18%
82%
Intr.

Train AUC = 0.76
Test AUC = 0.73
Local
Env.
28%
15%
57%
Intr.

Train AUC = 0.79
Test AUC = 0.80
Local (0%)
Intr.
8%
92%
Env.

Train AUC = 0.84
Test AUC = 0.82
Local
Intr.
8%
32%
60%
Env.